

Eric Sclafani

FULL STACK DEVELOPER

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Experience

Junior Full Stack .NET Developer

Brooklyn, NY

KINGS COUNTY DISTRICT ATTORNEY'S OFFICE

April 2024 - Present

- Modernized and centralized data management systems to optimize essential city legal services across two full stack web applications, improving the ability for non-technical Legal staff to utilize data for decision-making, and eliminating duplicative steps in the data entry pipeline (CRUD). Transformed business requirements supplied by Legal staff into these applications.
- Streamlined internal IT operations by creating full stack web applications that interface directly with important SQL Server lookup tables, greatly simplifying the database maintenance process.
- Used Angular, Typescript, RxJS and HTML to develop intuitive front-end interfaces with custom components, reactive forms, form validation rules, and asynchronous observable-based event handling. Leveraged CSS, Bootstrap, and Angular Material to create modern-looking and responsive web designs.
- Designed backend ASP.NET Core RESTful API endpoints to handle HTTP requests that execute T-SQL stored procedures and upload files to network drives. Created SQL Server relational database tables and developed custom T-SQL stored procedures. Established special Windows folder permission rules to protect sensitive bureau data.
- Created SSRS reports hosted on a separate server that accept user-provided criteria and generated tabular data reports. Embedded reports into application frontends via HTML iframes.
- Wrote extensive detailed software documentation for multiple full stack web applications
- Resolved user-submitted tickets in timely and efficient manner via ServiceDesk software.

Research Assistant

Stony Brook, NY

U.S. GOVERNMENT RESEARCH PROGRAM: IARPA HIATUS

October 2022 - August 2023

- Worked on a federal research project funded by the Intelligence Advanced Research Projects Activity (IARPA) called HIATUS, which focuses on authorship attribution, the task of automatically identifying the author of a document.
- Developed Gram2Vec, a Python package utilizing spaCy and Pandas to convert text into vectors based on writing style. Differs from other embedding algorithms like Word2Vec in that each vector position corresponds to a concrete feature pertaining to grammar usage. This makes the vectors interpretable, a property that traditional word vectors often lack.
- Performed authorship attribution experiments with Gram2Vec using custom built K-Nearest Neighbors testing algorithm, as well as an evaluation suite provided by IARPA.
- Collaborated with team members to compile monthly progress reports.
- Research Paper: Peter Zeng, **Eric Sclafani**, Owen Rambow, "[Gram2Vec: An Interpretable Document Vectorizer](#)"

Skills

Frontend Angular, RxJS, TypeScript, JavaScript, HTML, jQuery, VueJS

UI CSS, SCSS, Angular Material, Bootstrap

Backend ASP.NET Core, .NET, C#, MVC, Python

Database SQL Server, SSRS, T-SQL, SQLite

Tools Git, Visual Studio, Linux

Other Pandas, Scikit-Learn, NumPy, spaCy, Plotly, Matplotlib

Education

M.A. Computational Linguistics

Stony Brook, NY

STONY BROOK UNIVERSITY

August 2021 - May 2023

B.A. Linguistics

Stony Brook, NY

STONY BROOK UNIVERSITY

August 2019 - May 2021